

- RAG Learning

2.2 Financial Planning

Business — Edexcel (9BS0) — Theme 2: Managing business activities

Financial planning is about looking ahead: forecasting sales, understanding how costs behave as output changes, calculating how much must be sold to break even, and setting budgets to control spending. It is the most calculation-heavy topic in Theme 2 and regularly supplies 4-mark "calculate" questions on Paper 2 (revenue, costs, contribution, break-even, margin of safety, variances). The evaluation skill is to recognise that every one of these tools rests on **forecasts and assumptions** — a break-even point is only as reliable as the price and cost figures that go into it.

Sales forecast — a prediction of future sales volume or value over a given period.

Sales volume — the number of units sold.

Sales revenue — the value of sales: selling price $\times$ quantity sold.

Fixed costs — costs that do not change with the level of output in the short run (rent, salaries, insurance).

Variable costs — costs that change in direct proportion to output (raw materials, packaging, piece-rate wages).

Contribution — the amount each unit sold contributes towards fixed costs and then profit: selling price – variable cost per unit.

Break-even point — the level of output at which total revenue equals total costs, so the business makes neither profit nor loss.

Margin of safety — the amount by which actual (or forecast) sales exceed break-even sales.

Budget — a financial plan for the future, setting targets for revenue and/or expenditure over a period.

Variance — the difference between a budgeted figure and the actual figure.

2.2.1 Sales forecasting

a) Purpose of sales forecasts

- **Production and operations planning** — how much to produce, stock to hold and capacity required.
- **Staffing (workforce planning)** — how many staff to recruit, when to take on seasonal workers.
- **Financial planning** — the basis of cash-flow forecasts, budgets and break-even analysis; supports applications for finance.
- **Marketing** — setting targets and planning promotions.

Methods range from extrapolating past trends (moving averages, covered quantitatively in 3.3) to market research and expert opinion.

b) Factors affecting sales forecasts

- **Consumer trends** — changes in tastes and fashions (e.g. the shift to plant-based foods, the rise of weight-loss drugs affecting snack sales), which may not continue past patterns.
- **Economic variables** — incomes, inflation, interest rates, unemployment, exchange rates, the business cycle. A product with a high YED will be especially sensitive (link to 1.2.5 and 2.5.1).
- **Actions of competitors** — new entrants, price cuts, new products or promotional campaigns can take sales away unexpectedly.

c) Difficulties of sales forecasting

- The future is uncertain; past data may not predict the future, especially in dynamic markets.
- New businesses and new products have no past data to use.
- External shocks (a pandemic, a heatwave, a supply crisis) cannot be predicted.
- Research used for forecasting may be biased or based on small samples.
- Over-optimism, particularly from entrepreneurs seeking finance.

Consequences of an inaccurate forecast: overestimates lead to excess stock, wasted production, unnecessary staff and cash-flow problems; underestimates lead to stock-outs, lost sales and disappointed customers.

2.2.2 Sales, revenue and costs

a) Sales volume and sales revenue

$\text{Sales revenue} = \text{Selling price} \times \text{Quantity sold}$ $\text{Sales volume} = \text{Sales revenue} \div \text{Selling price}$

Worked example: a bakery sells 4,800 loaves a month at £2.50. Revenue = $4,800 \times 2.50 =$ **£12,000**. If monthly revenue from cakes is £6,300 at £3.50 each, volume = $6,300 \div 3.50 =$ **1,800 cakes**.

b) Fixed and variable costs

$\text{Total variable costs} = \text{Variable cost per unit} \times \text{Output}$

$\text{Total costs} = \text{Fixed costs} + \text{Total variable costs}$

$\text{Profit} = \text{Total revenue} - \text{Total costs}$

Worked example: the bakery's fixed costs are £5,500 a month; each loaf costs £0.90 in variable costs. At 4,800 loaves: TVC = $4,800 \times 0.90 =$ £4,320; TC = $5,500 + 4,320 =$ £9,820; profit = $12,000 - 9,820 =$ **£2,180**.

Some costs are **semi-variable** (e.g. a phone bill with a fixed line rental plus call charges). Fixed costs are fixed only in the short run and only within a range of output: expanding beyond current capacity requires new premises, creating a step in fixed costs. **Average (unit) cost** = total cost $\div$ output = $9,820 \div 4,800 =$ £2.05 per loaf; it falls as output rises because fixed costs are spread over more units.

2.2.3 Break-even

a) Contribution

$\text{Contribution per unit} = \text{Selling price} - \text{Variable cost per unit}$

$\text{Total contribution} = \text{Contribution per unit} \times \text{Units sold}$ (or total revenue – total variable costs)

$\text{Profit} = \text{Total contribution} - \text{Fixed costs}$

b) and c) Break-even point

At break-even: $\text{Total fixed costs} + \text{Total variable costs} = \text{Total revenue}$

Using contribution: $\text{Break-even output} = \text{Fixed costs} \div \text{Contribution per unit}$

Worked example. A company makes phone cases. Selling price £25; variable cost £10 per unit; fixed costs £30,000 per month; forecast sales 2,600 units per month.

- Contribution per unit = $25 - 10 = \text{£15}$
- Break-even = $30,000 \div 15 = \text{2,000 units}$ (break-even revenue = $2,000 \times \text{£25} = \text{£50,000}$)
- Check: at 2,000 units, TR = £50,000; TC = $30,000 + (2,000 \times 10) = \text{£50,000}$ ✓
- Profit at 2,600 units = $(2,600 \times 15) - 30,000 = 39,000 - 30,000 = \text{£9,000}$

What-if calculations:

- **Price rise to £28:** contribution = £18; break-even = $30,000 \div 18 = 1,666.7$, so **1,667 units** (always round *up* — 1,666 units would still make a small loss).
- **Variable cost rises to £12:** contribution = £13; break-even = $30,000 \div 13 = 2,307.7$, so **2,308 units**.
- **Target profit of £12,000:** required output = $(\text{fixed costs} + \text{target profit}) \div \text{contribution} = (30,000 + 12,000) \div 15 = \text{2,800 units}$.

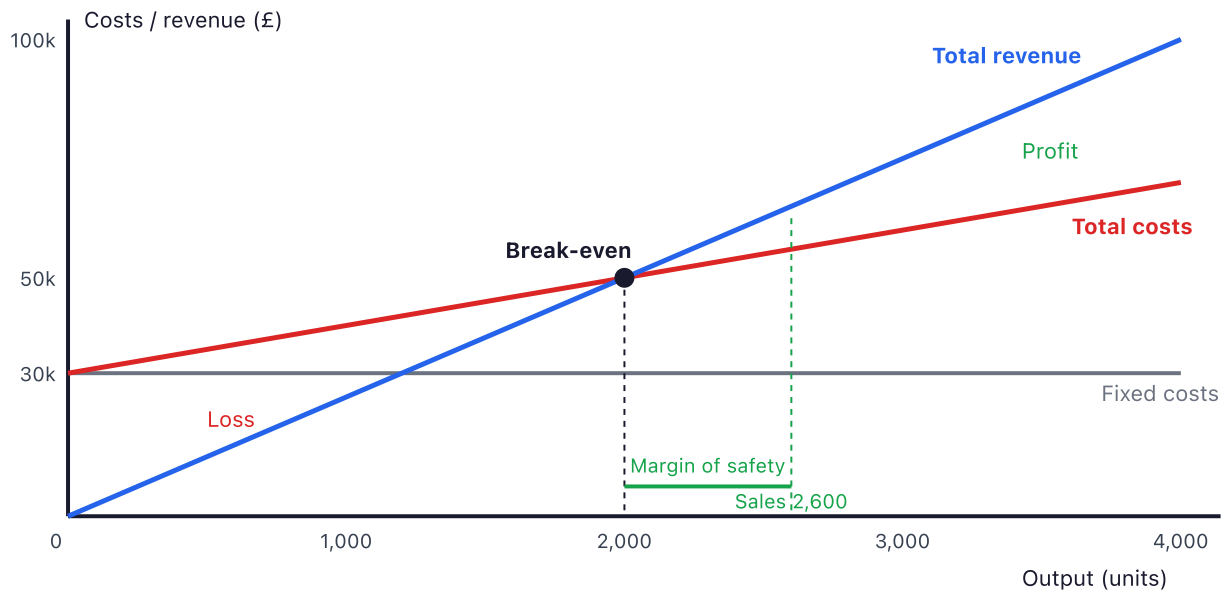
d) Margin of safety

$\text{Margin of safety} = \text{Actual (or forecast) sales} - \text{Break-even sales}$

In the example: $2,600 - 2,000 = \text{600 units}$ (or £15,000 of revenue). As a percentage of sales: $600 \div 2,600 \times 100 = \text{23.1\%}$ — sales could fall by 23.1% before the business made a loss. Note that profit = margin of safety $\times$ contribution per unit ($600 \times \text{£15} = \text{£9,000}$).

e) Interpretation of break-even charts

Break-even chart: phone cases (per month)



Reading the chart: the **fixed-cost line** is horizontal; the **total-cost line** starts at the fixed-cost level (at zero output, total cost = fixed cost) and rises with variable costs; the **total-revenue line** starts at the origin. Where TR crosses TC is the break-even point. To the left the vertical gap between TC and TR is the **loss**; to the right it is the **profit**. The horizontal distance from break-even to actual sales is the **margin of safety**.

Effects of changes on the chart: a price rise makes the TR line steeper, so break-even falls; a rise in variable cost makes the TC line steeper, so break-even rises; a rise in fixed costs shifts the FC and TC lines up in parallel, so break-even rises.

f) Limitations of break-even analysis

- Assumes all output is sold and there is no stock build-up.
- Assumes a single selling price, whereas firms discount and sell at different prices to different customers.
- Assumes variable costs per unit are constant; in reality bulk-buying discounts reduce them and overtime raises them.
- Assumes fixed costs are fixed at all output levels, ignoring step changes.
- Difficult to use for firms selling many products, where fixed costs must be allocated between products.
- Only as good as the data: forecasts of sales, prices and costs may be wrong.

- A static model: it shows a single point in time in a dynamic market.

Value despite limitations: it is quick and simple, useful for “what if” scenarios, an essential part of a business plan, and it shows how risky a venture is through the margin of safety.

Exam Tip: Round break-even output **up** to the next whole unit. Always state units (“2,000 phone cases per month”), and for a margin-of-safety question give the figure in units and, if useful, as a percentage. In evaluation, a small margin of safety is the key sign of vulnerability — link it to the context (e.g. a new competitor opening nearby).

2.2.4 Budgets

a) Purpose of budgets

- **Planning** — forces managers to think ahead about revenue and spending.
- **Control** — actual performance can be compared with the budget and corrective action taken.
- **Coordination** — ensures departments work towards the same objectives and do not overspend in total.
- **Motivation** — budget holders have targets and responsibility (delegated budgets — links to Herzberg).
- **Allocation of resources** — directs money to priorities.
- **Accountability** — managers are responsible for their budgets.

Types include **income (revenue) budgets**, **expenditure (cost) budgets** and **profit budgets** (income minus expenditure).

b) Types of budget

	Historical figures (incremental)	Zero-based
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Method	Based on last year's budget, adjusted (e.g. +3% for inflation)	Every budget starts from zero; each item of spending must be justified afresh
Advantages	Quick, simple, cheap; based on real past experience; stable	Eliminates wasteful spending carried over from past years; resources allocated to current priorities; managers must think about value for money
Disadvantages	Builds in past inefficiencies; ignores changes in the market; encourages "spend it or lose it" behaviour	Time-consuming and costly; requires skilled managers; may cause conflict between departments; hard for activities whose benefits are hard to measure (e.g. training)

c) Variance analysis

Variance = Actual figure – Budgeted figure

- **Favourable (F)** — the variance increases profit: revenue higher than budgeted, or costs lower than budgeted.
- **Adverse (A)** — the variance reduces profit: revenue lower than budgeted, or costs higher than budgeted.

Item (£, one month)	Budget	Actual	Variance
Sales revenue	50,000	46,000	4,000 A
Materials	20,000	18,500	1,500 F
Wages	12,000	12,800	800 A
Marketing	3,000	3,000	0
Total costs	35,000	34,300	700 F
Profit	15,000	11,700	3,300 A

Interpreting it: the profit shortfall comes mainly from lower sales. The favourable materials variance is not necessarily good news — it may simply reflect lower output because sales were lower (or cheaper, lower-quality materials). The adverse wages variance despite lower sales needs investigating (overtime? a pay rise?). A variance tells managers *where* to look, not *why* it happened.

Variances can also be expressed as a percentage of budget: revenue $4,000 \div 50,000 \times 100 = 8\%$ adverse.

d) Difficulties of budgeting

- Budgets are based on forecasts that may be inaccurate, especially in dynamic markets or for new businesses.
- External changes (inflation, as in 2022–23 when energy and material costs rose sharply) can make budgets unrealistic quickly.
- Inflexible budgets may prevent managers responding to opportunities.
- Budget setting takes time; imposed budgets demotivate, while negotiated budgets may be padded ("budgetary slack").
- Departments may compete for resources, and managers may spend unused budget at year-end to avoid a cut next year.
- Focus on hitting short-term budget targets can damage long-term investment.

Common mistakes

- Calculating break-even as fixed costs $\div$ selling price (it is $\div$ contribution).
- Rounding break-even down instead of up.
- Labelling a lower-than-budgeted cost as adverse — lower costs are favourable.
- Assuming a favourable variance is always good news.
- Confusing contribution with profit: contribution must first cover fixed costs.
- Drawing the TC line from the origin rather than from the fixed-cost level.

Quick recall

- Sales forecasts: plan production, staff, finance; affected by consumer trends, economic variables, competitors; hard to get right
- Revenue = price $\times$ quantity; TC = FC + (VC per unit $\times$ output)
- Contribution per unit = price – variable cost per unit
- Break-even = fixed costs $\div$ contribution per unit (round up)
- Margin of safety = actual sales – break-even sales; profit = MoS $\times$ contribution per unit
- Required output for a target profit = (FC + target profit) $\div$ contribution
- Break-even limitations: single price, constant VC, all output sold, fixed FC, forecasts
- Budgets: plan, control, coordinate, motivate; historical versus zero-based
- Variance = actual – budget; favourable raises profit, adverse lowers it